Developing nano-enabled carriers for efficient delivery of Oxytetracycline and Streptomycin to prevent Huanglongbing disease in citrus plants

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I. Abstract

The citrus disease Huanglongbing (HLB), commonly referred to as citrus greening, is one of the most serious threats to the citrus industry in the United States. Current methods, primarily using trunk injection and root drench with solutions of antibiotics, are not effective in treating this disease. The purpose of this research is to investigate how novel engineered nanocarriers containing antibiotics to treat this disease move throughout citrus plants when applied foliarly. We will be using nanocarriers made using a flash nanoprecipitation method to deliver Oxytetracycline and Streptomycin (our chosen antibiotics) to the phloem and roots of valencia orange trees with high efficiency. An effective method of delivering these antibiotics to treat HLB has not been developed. Due to the damage that this disease does to the citrus industry each year, it is imperative that a treatment is found before all citrus plants in the US are affected. My experimental design will entail dosing different volumes of five different nanocarrier suspension onto the leaves of the valencia orange trees, and after three days, measuring the amount of those particles in different parts of the plant using ICP-MS. After these initial tests, experiments will be repeated with the best performing particles to confirm the highest levels of antibiotic delivered into the roots and through the phloem (the vascular tissue in plants) using an ELISA assay to detect antibiotic levels.

II. Objectives and Contribution to the Research

The main objective of this research is part of a larger project to find a treatment for Huanglongbing (HLB) disease. HLB, which is caused by *Candidatus* Liberibacter asiaticus (*C*Las), is a serious citrus disease that has had significant negative impacts on the citrus industry in the United States, with Florida suffering from a 74% decline in production and a loss of $1 billion annually (Li et al 2020). Of the major citrus producing states, Florida is the most affected with nearly every commercial citrus tree being infected with this disease. In Texas, the rate is about 80% (Sétamou 2020), while in California, commercial trees are still largely unaffected but early detection and removal is critical to keep it this way (Yuan et al 2020). *C*Las is spread through the Asian citrus psyllid, so treatments in the past have included aggressive insecticide application, but there are environmental concerns about insecticide resistance (Wood and Goulson 2017). Similar concerns have been raised with large scale foliar antibiotic spray, which has also proved to be ineffective (McKenna 2019). This research will improve the efficiency of these antibiotics by increasing their uptake into leaves and phloem, the site of the pathogen, and will reduce the amount of antibiotics needed to keep the plants healthy.

The research question I will be investigating in this proposal is how the nanocarriers travel through citrus leaves and move throughout the tree vasculature, specifically targeting the phloem and roots. The nanoparticles applied to the leaves will be internalized into the valencia orange plant and travel into the phloem. Based on prior work with gold nanoparticles at Carnegie Mellon, we hypothesize that up to 1-10μM of the antibiotic in the particles will be found in the phloem, and expect to find the best Flash NanoPrecipitation (FNP) nanocarrier properties and application strategies to accomplish this. Preliminary data developed at Carnegie Mellon has shown this to be possible using the FNP method for production of nanoparticles. This is the same procedure used to make the Pfizer COVID-19 vaccine. It is cheap to produce and is highly scalable, but is untested in plants.

Currently, most of the experiments done with antibiotic application on citrus plants have been done using solutions of the antibiotic, through methods such as trunk injection, root drench, and leaf application. When done using a solution of antibiotics, trunk injection and root drench have proved to be the most effective in moving throughout the plant, but neither are ideal (Hu and Wang 2016). Trunk injection is costly and time consuming, while root drench is not viable in the field (Li et al 2020). The research I will be conducting will involve the testing of nanoparticles with hydrophilic surfaces containing antibiotics. They are expected to outperform the other options for antibiotic delivery into citrus plants

While previous research done by Dr. Lowry's lab at CMU has looked at the movement of nanoparticles with gold through plants; the usage of nanoparticles for antibiotic delivery via the phloem requires more investigation. This study would potentially increase the understanding of how antibiotics in nanoparticles move through citrus plants and can then be applied to other nanoparticles and other plants with similar physiology. The same principles can also be used to deliver herbicides, nutrients, and other plant protection products with high efficiency.

III. Methodology

Testing will be performed on valencia orange trees due to their commercial significance and availability. We will have 30 to 40 trees that will be available for testing and observation. These trees will be purchased commercially as saplings about 1-2 feet tall because that size is feasibly digestible. We will have a watering and lighting system set up in the lab able to sustain the trees throughout the summer. I will complete this part of the project before summer, so the plants will be ready for the proposed research. FNP particles with different hydrophilic shell materials will be tested to determine which shell material properties provide the best uptake.

Application of the nanoparticles will take place in individual trees throughout the research period. Candidate nanocarriers include those with a hydrophobic core (with the Gd tracer) to hold the antibiotic, and a hydrophilic shell of naturally-occurring compounds including fish gelatin, soybean lecithin, and semisynthetic hydroxypropyl methylcellulose acetate succinate (HPMCAS). These shell materials are biodegradable and highly promising candidates. Different volumes of the nanocarrier suspension (1 g/L) will be applied foliarly to the five youngest leaves of the saplings. After three days, the saplings will then be cut into sections to determine the location of the particles. Each particle also contains gadolinium that cannot leach from the particle. We use this gadolinium as a tracer to find the particles in the plant. The different plant sections will include the dosed leaves, younger leaves, older leaves, trunk, and roots. The plant sections will be digested with hydrogen peroxide and nitric acid, heated, filtered, and then the Gd levels will be measured using induced coupled plasma mass spectrometry (ICP-MS). Presence of the FNP particles will also be confirmed using single particle inductively coupled plasma time of flight mass spectrometry (spICP-TOF-MS) and confocal microscopy. The spICP-TOF-MS is a unique instrument (<5 in the world are used for environmental research) that enables quantitative assessment of the number of particles present in the different plant tissues. Tests will be replicated on six plants to provide statistical relevance to the results. Once the locations of the particles are confirmed, we will use enzyme-linked immunoassay (ELISA) to detect the presence of antibiotics throughout the plant phloem. Our goal is to get 1-10μM antibiotics into the phloem.

Traditional ICP-MS is used to measure trace metal elements (Gd in the particles) in biological fluids. This has already been used in previous work done by the research team at Carnegie Mellon and has proved to be an effective way to track nanoparticles through plants. ELISA is a diagnostic assay that can detect organic compounds undetectable by ICP-MS and will be used to determine and confirm the concentrations of Oxytetracycline and Streptomycin.

The first half of this project will be dedicated to finding the optimal volume and application strategy of the nanocarrier suspension for five different particle types, mainly using ICP-MS to detect the tracer metal. The second half will focus more on using the ELISA assay to determine the concentration of antibiotic throughout the plants, using the optimal methods determined in the first half of the summer. By the end of the summer, I would like to demonstrate that several of the particles can successfully travel to the phloem and deliver the required dose of antibiotics.

IV. Background

My faculty mentor is Dr. Greg Lowry in the Civil and Environmental Engineering Department. I met Dr. Lowry through the Environmental Chemistry and Thermodynamics class he taught in Fall 2022. This class is offered through the CEE department and required for my major. I really enjoyed the content and his teaching and enthusiasm, so when he mentioned the need for undergraduate research assistants in his lab, I was very excited for the opportunity. I began attending lab meetings with Dr. Lowry's research group at the end of Fall 2022 and began actively working in the lab in January 2023. This proposal is a continuation of the work I have already been doing this semester, so I feel well prepared to build on it over the summer. Classes that have prepared me for this include Environmental Chemistry and Thermodynamics, Modern Biology, and Environmental Engineering.

V. Feedback and Evaluation

Dr. Lowry will provide feedback on the project in addition to Dr. Hagai Kohay, a postdoctoral researcher. We will meet weekly during the summer. This research is part of an overall project being conducted in the CEE department, so I will also be presenting my progress to the larger plant team at biweekly meetings, and once during the summer to Dr. Lowry's whole research group. Other team members include PhD students Valeria Nava and Ben Therrien. Both are working on plant nanobiotechnology projects and are invested in the success of this project. I will keep a lab notebook with all of my findings and observations.

VI. Dissemination of Knowledge

The results of my research will be shared at the Meeting of the Minds research symposium in May 2024. In addition to that, there will likely be a poster session hosted at the end of the summer by the Civil and Environmental Engineering department. My final report will take the form of a poster.

Additionally, since this project is part of a larger project being worked on by not only Carnegie Mellon University, but also several other universities, my research will likely be used in a publication in the next one to two years, and I plan to be a coauthor on that paper. The methods we develop over the summer will ideally end up being used in field testing in Florida in spring 2024 that will hopefully result in an effective treatment for HLB.

References

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